Bulk–Boundary Correspondence for Stabilizer States
and the Resolution of the 2D Classification Conjecture

Bowen Yang

Harvard University

March 11, 2026

Based on: B. Ruba and B. Y., *Witt Groups and Bulk–Boundary Correspondence for Stabilizer States*, arXiv:2509.10418 (2025).

Talk goals

- Explain a clean bulk–boundary correspondence (BBC) for translation-invariant (TI) stabilizer states using algebra over Laurent polynomial rings.
- Show how BBC gives the missing ingredient to resolve the 2D TI stabilizer-state classification conjecture.

Main references: arXiv:2509.10418 and earlier work cited therein.

Motivation: classify TI stabilizer phases

- Stabilizer states are unique invariant states of commuting unitaries built from maximal abelian subgroups of (generalized) Pauli operators.
- In 2D, they realize every abelian anyon model that admits a gappable boundary. The construction relies on non-prime qudits.
- An exact correspondence was conjectured.
- We will see how the boundary algebra in 1D controls the bulk anyon data in 2D, settling the conjecture.
- The bulk-boundary correspondence holds for higher dimensional TI stabilizer codes leading to higher dimensional results.

Laurent-polynomial method (review)

- Fix spatial dimension d and on-site $\mathbb{Z}_n$ degrees of freedom.
- Work over the Laurent polynomial ring

$$R = \mathbb{Z}_n[x_1^{\pm 1}, \ldots, x_d^{\pm 1}], \qquad x_i \mapsto x_i^{-1} \text{ (involution)}.$$

- Canonical *free* R-module with standard anti-hermitian pairing

$$P = L_0 \oplus L_0^* \simeq R^{2q}, \quad \Omega = \begin{bmatrix} 0 & I_q \\ -I_q & 0 \end{bmatrix},$$

  encoding Pauli commutation and translations.
- A TI stabilizer *state* is specified by a Lagrangian submodule $L \subset P$ with $L = L^{\perp}$.

Charges via homological algebra

- **Charge modules** (localized excitations and detectors) are

$$Q_i := \mathrm{Ext}_R^{i+1}(P/L,\, R), \qquad D_i := \mathrm{Tor}_{i+1}^R(P/L,\, \tilde{R}) \simeq \mathrm{Hom}_{\mathbb{Z}_n}(Q_i, \mathbb{Z}_n).$$

- In $d = 2$ one has $Q_i = 0$ for $i > 0$; the bulk anyon group is Q_0.
- It is conjectured that Q_0 is finite, which we proved[1] using properties of Gorenstein rings.
- The finite abelian group Q_0 is the candidate for an abelian anyon theory.
- If L were a direct summand, all Q_i would vanish (trivial phase).
- There is a natural quadratic form $\theta : Q_0 \to \mathbb{C}^*$, indicating the spin of each abelian anyon.

[1] See Homological Invariants of Pauli Stabilizer Codes. Commun. Math. Phys. 405, 126 (2024).

Setup: half-space and boundary algebra

Half-space cut. Fix a unit vector $\hat{n}$ and consider the half-space

$$H_{\hat{n}} = \{\, \mathrm{r} \in \mathbb{Z}^d \mid \mathrm{r} \cdot \hat{n} \geq 0 \,\}.$$

A translation-invariant (TI) stabilizer state with Lagrangian $L \subset P \simeq R^{2q}$ (over $R = \mathbb{Z}_n[x_1^{\pm 1}, \ldots, x_d^{\pm 1}]$) induces *boundary* degrees of freedom supported near $\partial H_{\hat{n}}$.

Boundary ring. Choosing coordinates so the boundary is orthogonal to x_d, write

$$R_\partial = \mathbb{Z}_n[x_1^{\pm 1}, \ldots, x_{d-1}^{\pm 1}].$$

Bulk translation by x_d becomes a *shift* transverse to the boundary and is forgotten at the boundary, leaving R_∂-linearity.

Goal. Package all boundary observables and commutation relations into a single R_∂-module with a canonical pairing.

Definition: boundary operator module $(P_\partial, \Omega_\partial)$

Construction (informal).

1. Restrict Pauli operators in $H_{\hat{n}}$ to those commuting with the stabilizers supported strictly in $H_{\hat{n}}$ and mod out by these stabilizers.
2. Take the induced R_∂-action from translations *parallel* to the boundary.
3. Push forward the bulk anti-hermitian commutator pairing Ω to a sesquilinear pairing Ω_∂ on the quotient.

This yields an R_∂-module P_∂ with

$$\Omega_\partial : P_\partial \times P_\partial \to R_\partial,$$

sesquilinear with respect to the involution $x_i \mapsto x_i^{-1}$, $1 \leq i \leq d-1$.

Definition. The pair $(P_\partial, \Omega_\partial)$ is the *boundary operator module*. It is so-called quasi-symplectic by our terminology.

Quasi-symplectic structure & Witt viewpoint

Quasi-symplectic. Unlike the bulk free module $P \simeq R^{2q}$, the boundary module P_∂ need not be projective. The pairing Ω_∂ induces an inclusion, instead of an isomorphism

$$P_\partial \to P_\partial^* := \mathrm{Hom}_{R_\partial}(P_\partial, R_\partial),$$

i.e. $(P_\partial, \Omega_\partial)$ is *quasi-symplectic*.

Metabolic pieces. A submodule $L_\partial \subset P_\partial$ is *Lagrangian* if $L_\partial = L_\partial^\perp$ (with respect to Ω_∂). A module admitting such an L_∂ is called *metabolic* (Witt-trivial).

Witt group. The *Witt group* $W(R_\partial)$ of quasi-symplectic modules identifies $(P_\partial, \Omega_\partial)$ up to adding/removing metabolic summands:

$$(P_\partial, \Omega_\partial) \sim (P_\partial, \Omega_\partial) \oplus (\text{metabolic}).$$

This lets us test for gappable boundary without fully classifying P_∂.

Why Witt groups?

- Classifying quasi-symplectic boundary data up to addition/removal of *metabolic* pieces is captured by a Witt group $W(R_\partial)$.
- Differences from classical L-theory: no projectivity or unimodularity assumed.
- Advantage: we can decide whether a quasi-symplectic module admits a Lagrangian without classifying it completely.

Boundary charge module E_L and the BBC map

From boundary operators to charges. Given the bulk Lagrangian $L \subset P$ and a boundary cut, there is a canonically associated finite R_∂-module $E_L = P_\partial^*/P_\partial$ (the *boundary charge module*), extracted from $(P_\partial, \Omega_\partial)$.

Bulk charges. In the Laurent-polynomial framework, bulk excitations in $d = 2$ are captured by

$$Q_0 := \mathrm{Ext}_R^1(P/L, R),$$

equipped with its quadratic refinement θ (topological spin).

Boundary-to-bulk map.

BBC in algebra

There is a canonical homomorphism

$$V : E_L \longrightarrow Q_0,$$

relating boundary charges to bulk anyons.

In $d = 2$, V is an *isomorphism* and intertwines the quadratic forms (boundary q equals $\theta \circ V$).
In $d > 2$, V is an isomorphism only for a "generic" half geometry.

1D boundary: quadratic form from quasi-symplectic data

We need to look at the Witt group $W(R_\partial)$ for $R_\partial = \mathbb{Z}_n[x, x^{-1}]$.

- In 1D, to any quasi-symplectic $(P_\partial, \Omega_\partial)$ one associates a finite abelian group

$$E_{P_\partial} := P_\partial^* / P_\partial,$$

  carrying a natural nondegenerate quadratic form q.
- Structural facts in $d = 1$:
  1. Every symplectic module is *metabolic* (Witt-trivial).
  2. Every quasi-symplectic module becomes metabolic after coarse-graining.

Summary

Objects:

Bulk:	(P, Ω),	$L \subset P$ Lagrangian,	$Q_0 = \mathrm{Ext}^1_R(P/L, R)$, θ
Boundary:	$(P_\partial, \Omega_\partial)$ over R_∂,	E_L, q	

Maps: $V : E_L \to Q_0$, $q = \theta \circ V$ in $d = 2$.

Conclusions (2D):

- V is an isomorphism; (Q_0, θ) is nondegenerate because q (which is defined entirely algebraically) is nondegenerate.
- Lagrangian subgroup exists $\Rightarrow$ *gappable boundary*.
- TI stabilizer state $\Rightarrow$ abelian anyon model with gapped boundary.

Higher d and fractons (outlook)

- BBC exists in all d; for *generic* cuts, V is surjective and often an isomorphism.
- For special boundary orientations, mobility constraints may obstruct surjectivity (fracton-like behavior).
- Witt-group methods give a systematic handle on when boundary Lagrangians exist beyond 2D.

Abelian Anyon Theory (backup)

Metric groups. Let E be a finite abelian group and $q : E \to k^\times$ a quadratic form over an algebraically closed field k of characteristic 0.

$$b(a_1, a_2) = \frac{q(a_1 + a_2)}{q(a_1)q(a_2)}$$

is bilinear. The pair (E, q) is:

- a *pre-metric group*;
- a *metric group* if b is also non-degenerate.

Each $q(a)$ is a root of unity, and (E, q) decomposes as a direct sum of its p-primary components.

Categorical meaning.

$$\{\text{pointed braided fusion categories}\} \leftrightarrow \{\text{pre-metric groups}\},$$
$$\{\text{pointed modular tensor categories}\} \leftrightarrow \{\text{metric groups}\}.$$

Metric groups thus classify pointed modular tensor categories and corresponding (2+1)-dimensional TQFTs.